SERANGOON JUNIOR COLLEGE
2008 JC2 PRELIMINARY EXAMINATIONS

MATHEMATICS

Higher 2 **9740/Paper 1**

Thursday **21 August 2008**

Additional materials: Writing paper

List of Formulae (MF15)

TIME : 3 hours

READ THESE INSTRUCTIONS FIRST

Write your name and class on the cover page and on all the work you hand in.
Write in dark or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets **[ ]** at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

This question paper consists of 7 printed pages and 1 blank page.

Answer all questions (100 marks).

1 The first term of an arithmetic series is 3 and the first term of a geometric series is 4. The arithmetic and geometric series have both b as the common difference and common ratio respectively. The sixth term of the arithmetic series is equal to the sum of the third and fourth terms of the geometric series. Find all the possible values of b. For convergence in the geometric series, find its sum to infinity.
[Leave all your answers in exact form] **[5]**

2 Solve the inequality

$$|x+1| > x^3 + 2x^2 - 5x - 6 .$$

Hence solve $\big||x|+1\big| > |x|^3 + 2x^2 - 5|x| - 6$. **[5]**

3 Functions f and g are defined by

$$\mathrm{f} : x \mapsto \sqrt{x} \text{ for } x > 0,$$
$$\mathrm{g} : x \mapsto \sin^2 x \text{ for } -\pi \le x \le 0$$

(i) Show that the composite function fg does not exist. **[2]**

(ii) Find the largest domain of g such that the composite function fg exists. **[1]**

(iii) Using the restricted domain found in part **(ii)**, find the composite function fg in a similar form. Find also the range of fg. **[3]**

4 Sketch in an Argand diagram, the set of points representing all complex numbers z satisfying the following inequalities

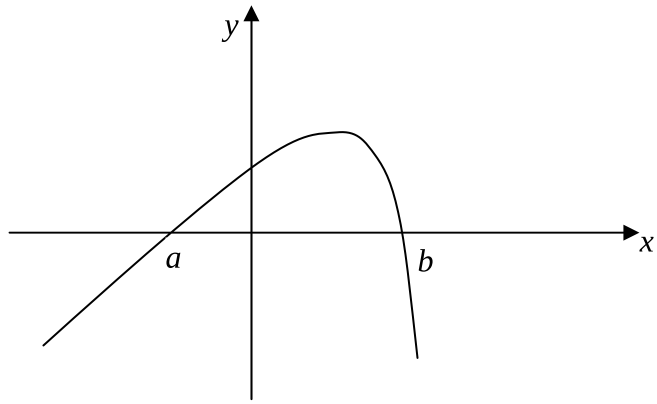
$$0 \le \arg(z-1) \le \frac{\pi}{4} \quad \text{and} \quad |z+1-i| \le |z-5-i|.$$ **[3]**

Hence, find

(i) the maximum value of $\arg(z)$, **[1]**

(ii) the exact range of values of $|z|$. **[2]**

5 The diagram shows the graph of $y = 3x - 2^x + 3$. The 2 roots of the equation $3x - 2^x + 3 = 0$ are denoted by a and b where a < 0 and b > 0.



(i) Find the values of a and b, each correct to 3 decimal places. **[2]**

The real numbers x_n satisfies the recurrence relation,

$$x_{n+1} = \frac{1}{3}(2^{x_n}) - 1 \qquad \text{for } n \ge 1.$$

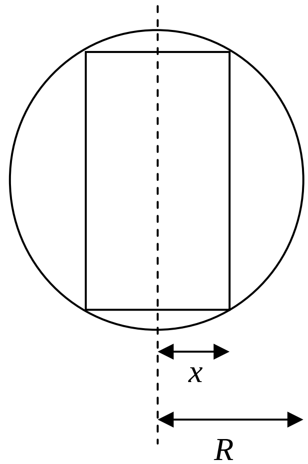
(ii) Using the result in **(i)**, show that if the sequence converges, it will converge to either a or b. **[2]**

(iii) By considering $x_n - x_{n+1}$, prove that

$$x_n > x_{n+1} \text{ if } a < x_n < b, \quad x_n < x_{n+1} \text{ if } x_n < a \text{ or } x_n > b.$$ **[2]**

[Turn Over

6 The diagram below shows the cross-section of a cylinder of radius x that is inscribed in a sphere of fixed internal radius R. Show that $A^2 = 16\pi^2 x^2 (R^2 - x^2)$, where A is the curved surface area of the cylinder. Prove that, as x varies, the maximum value of A is obtained when the height of the cylinder is equal to its diameter.



[8]

7 Let $\mathrm{f}(r) = \dfrac{\sin\left[(2r+1)\theta\right]}{\cos\theta}$, $r \in \mathbb{Z}^+$.

(i) Show that $\mathrm{f}(r) - \mathrm{f}(r-1) = \mathrm{A}\cos(\mathrm{B}r\theta)\tan(\theta)$, where A and B are to be determined. **[3]**

(ii) By using the result in part **(i)**, show that

$$\cos 2\theta + \cos 4\theta + \ldots + \cos 2N\theta = \frac{1}{2}\left[\frac{\sin\left[(2N+1)\theta\right]}{\sin\theta} - 1\right].$$ **[5]**

8 **(a)** Express $(3+4x)^{\frac{1}{2}}$ as a series of descending powers of x, up to and including the third non-zero term, simplifying the coefficients. State the set of values of x for which the series expansion is valid. **[4]**

(b) Given that $y = \dfrac{\sin x + \cos x}{\sin x - \cos x}$, show that $\dfrac{dy}{dx} + y^2 + 1 = 0$.

Hence, find the Maclaurin's series for y, up to and including the term in x^4. **[5]**

9 The parametric equations of a curve are given by

$$x = \frac{1}{4}\left(e^{2t} - 2t\right) \text{ and } y = e^t \text{ where } 0 \le t \le 1.$$

(i) Find $\dfrac{dy}{dx}$ and show that $\dfrac{d^2y}{dx^2} < 0$ for $0 < t \le 1$ **[5]**

(ii) Sketch the curve and find the value of the area bounded by the curve, the lines $x = \dfrac{1}{4}\left(e^2 - 2\right)$ and $y = 1$. **[5]**

10 **(a)** Find $\displaystyle\int \frac{e^{\sqrt{x}}}{\sqrt{x}}\,dx$. **[3]**

(b) Use the substitution $t = \sqrt{x^2+1}$ to find $\displaystyle\int \frac{1}{3x\sqrt{x^2+1}}\,dx$. **[4]**

(c) Show that $\displaystyle\int (x+2)^2 \ln(x+2)\,dx = \frac{1}{9}(x+2)^3\left[3\ln(x+2) - 1\right] + c$. Hence find the exact value of $\displaystyle\int_{-\frac{3}{2}}^{0} \left|(x+2)^2 \ln(x+2)\right|\,dx$. **[5]**

11 The curve C has the equation $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = \dfrac{ax^2 + bx + c}{x + d}$, where a, b, c and d are constant integer values such that $ad^2 - bd + c \neq 0$.

The line $y = 3x - 5$ is an asymptote to the curve C.

(i) By rewriting $\mathrm{f}(x) = Q(x) + \dfrac{k}{x + d}$ where $Q(x)$ is a polynomial of x and k is a constant, or otherwise, show that $a = 3$ and $b = 3d - 5$. **[3]**

(ii) Show that C has two stationary points if and only if $c > -5d$. **[3]**

(iii) Hence, using the minimum value of c and finding the value of b, sketch the curve when $d = -1$. Your sketch should include all asymptotes, stationary points and axial intercepts if any. **[4]**

(iv) Find the equations of asymptotes when C undergoes transformation from $\mathrm{f}(x)$ to $2\mathrm{f}(x - 3)$. **[2]**

12 **(a)** Relative to the origin O, position vectors of A, B and C are **a**, **b** and **c** respectively where **a**, **b** and **c** are non-parallel vectors. D is the mid-point of AB and E is the point of trisection of AC nearer to C. The line BE meets line DC at point F. Show that the position vector of F is $\frac{1}{4}(\mathbf{a}+\mathbf{b}+2\mathbf{c})$. **[5]**

(b) The plane π_1 has equation $\mathbf{r}\cdot\begin{pmatrix}-1\\2\\0\end{pmatrix}=4$. The line l passes through the points P and Q with position vectors $\begin{pmatrix}3\\11\\1\end{pmatrix}$ and $\begin{pmatrix}10\\7\\3\end{pmatrix}$ respectively.

(i) The foot of the perpendicular from P to the plane π_1 is given by the point F(6,5,1). Verify that Q is a point on π_1 and hence find a vector parallel to the reflection of l in π_1. **[4]**

(ii) The plane π_2 contains the line l and is perpendicular to π_1. Show that the equation of π_2 can be expressed as $\mathbf{r}\cdot\begin{pmatrix}2\\1\\-5\end{pmatrix}=12$. **[2]**

(iii) Given that the plane π_3 has equation $\mathbf{r}\cdot\begin{pmatrix}1\\3\\-5\end{pmatrix}=7$. Deduce the geometrical representation of the 3 planes, π_1, π_2 and π_3 by solving the systems of linear equations:

$$\begin{aligned}-x+2y&=4\\2x+y-5z&=12\\x+3y-5z&=7\end{aligned}$$

[2]

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